

# RingFlowUI: 3D Interface System using Ring-mounted Camera

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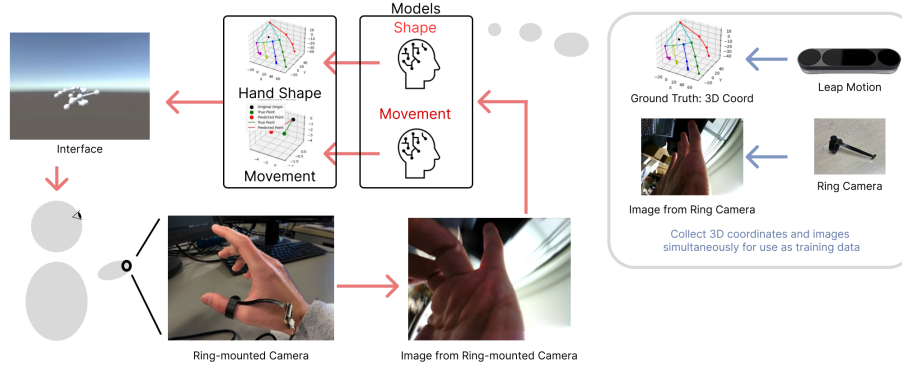


Figure 1: RingFlowUI architecture: ring camera, inference model, and interface pipeline.

## Abstract

We present RingFlowUI, a 3D interaction system using a finger-mounted ring camera. Our deep learning model infers hand shape and motion from videos captured by the ring camera, enabling natural 3D interaction without external tracking. Experimental results showed reasonable hand shape reconstruction, and moderately reliable motion estimation. We also demonstrated a 3D interaction system that leverages the proposed RingFlowUI, to map inferred hand motion into virtual space and to operate reliably even in regions typically invisible to HMD cameras, such as under a desk.

## CCS Concepts

• Human-centered computing → Pointing devices.

## Keywords

3D Interface, VR, AR, Deep Learning

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## 1 Introduction

Recent VR/AR headsets have promoted hand-based interaction as a natural way to manipulate virtual objects. While intuitive,

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they require raising arms up to virtual objects, causing fatigue and limiting reach. To reduce this burden, palm-level interfaces have been proposed [Chen et al. 2019; Han et al. 2022; Qi et al. 2023; Wu et al. 2020]. They enable relaxed interaction, but none achieves both comfort and usability, often trading one for the other.

We propose a new 3D interface using ring-mounted wide-angle camera, named RingFlowUI. The interface captures partial hand and background, and then estimates hand shape and motion with a deep-learning model. By training with ground-truth data from existing sensors, RingFlowUI can reconstruct hand shape from partial appearances and estimate motion from background cues. This enables natural 3D interaction without external tracking or direct visibility, while keeping the device compact.

## 2 Proposed Method

Figure 1 shows the architecture of RingFlowUI, consisting of three main components: Ring-mounted Camera, Hand Shape and Motion Estimation Model, and Interface System.

### 2.1 Ring-mounted Camera

We design a compact wide-angle camera module embedded in a 3D-printed ring worn on the user's finger. The ring-mounted form factor allows the camera to stay close to the hand. Because the device is attached to a finger, it can be used flexibly without being restricted to a fixed operating position. Its compact form factor also ensures comfortable and unobtrusive use during 3D interaction.

### 2.2 Hand Shape and Motion Estimation Model

As shown in Fig. 1, the ring-mounted camera captures only partial view of the hand. By leveraging deep learning models trained on Leap Motion data, RingFlowUI infers complete hand shapes

and motions from these partial observations, enabling accurate 3D interaction even with limited visual input.

For estimation, we use a dual-CNN, 2D-CNN for hand shape and a 3D-CNN for motion. As shown in Fig. 2, the shape network predicts 3D coordinates for 21 joints per frame, while the motion network processes  $N$  frames to estimate 3D translation and rotation (six degrees of freedom), converted to a  $4 \times 4$  transformation matrix via Rodrigues' formula for real-time pose updates.

Training requires 3D joints coordinates obtained from Leap Motion as ground-truth. The combined loss is defined as

$$\mathcal{L} = \lambda_s \text{MSE}(\hat{J}, J) + \lambda_t \text{MSE}(\hat{T}, T) + \lambda_r \text{AngErr}(\hat{R}, R),$$

where  $\hat{J}$  and  $J$  are predicted and ground-truth joints,  $\hat{T}$  and  $T$  the translations, and  $\hat{R}$  and  $R$  the rotations. MSE is applied to joints and translations, while AngErr measures the angular difference between rotations. The weights  $\lambda_s, \lambda_t, \lambda_r$  balance the terms.

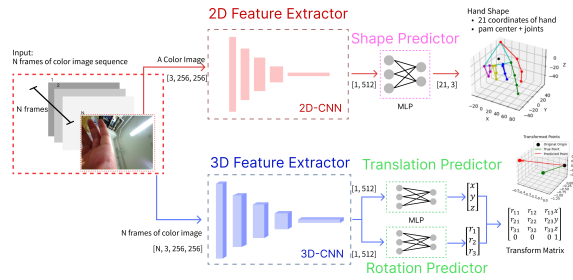


Figure 2: Dual CNN model for shape and motion estimation.

### 2.3 User Interface

The inferred hand shape and motion are mapped to a virtual hand model and further processed so that users can operate wide VR/AR spaces with minimal physical movement. This enables natural articulation and interaction even when the hand rests at the side or is partially occluded.

## 3 Experiment

We quantitatively evaluated our method using five-fold cross-validation. We used a dataset consisting of 25 video sequences (1,000 frames each at 30 fps) annotated 21 hand joints and inter-frame motion. The data was divided into 10-frame groups serve as the input of the model.

For each fold, 20 sequences were used for training and 5 for testing. Networks were trained for 20 epochs using the Adam optimizer ( $\text{lr} = 0.001$ ). Shape error was measured as the mean Euclidean distance of joint positions (mm), and motion error normalized by ground-truth translation magnitude.

Table 1 summarizes the results. Shape estimation achieves reasonable accuracy, while motion estimation shows a 37.6% error—above the 27% target derived from Fitts' law. This suggests that general motion is captured, but trajectory precision still needs improvement.

Figure 3 compares predicted and ground-truth hand shapes, confirming reliable reproduction of hand shape for natural interaction.

Table 1: Quantitative errors in shape and motion estimation from cross-validation experiments.

| Metric           | Mean | Top1 |
|------------------|------|------|
| Shape Error (mm) | 9.10 | 7.86 |
| Motion Error (%) | 37.6 | 32.2 |

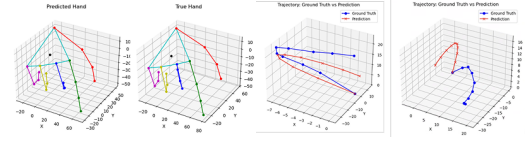


Figure 3: Comparison of predicted and Ground-truth of shape and movement

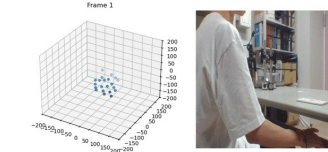


Figure 4: Free hand movement under the desk.

Figure 4 presents a 3D interaction demo where inferred hand motion is mapped into virtual space. Translation distance is scaled by movement speed, enabling both fine and large motions. The demo also shows that our method works in areas usually invisible to HMD cameras, such as under a desk, demonstrating the advantage of the ring-mounted viewpoint.

## 4 Conclusion

We proposed a ring-mounted camera system that infers hand shape and motion for natural 3D interaction. Experiments show it can reconstruct hands from limited input, though motion estimation still needs improvement.

We aim to enable intuitive 3D interaction through natural gestures in a relaxed posture. While not yet fully implemented, our results show its potential, and future work will refine motion estimation and mapping to virtual space for more precise control.

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